Morphological Classification of Galaxies in HST and JWST Images Using Classical DIP

Weekly Progress Report — Week 7

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ECE 501 – Digital Image Processing Course Project

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1 Purpose and Scope

Week 6 stabilized the segmentation/watershed algorithm and produced a fully verified CAS (Concentration, Asymmetry, Smoothness) feature table for all five Stephan's Quintet members, using real JWST NIRCam and HST ACS/WFC data. Week 7 takes that validated feature table and closes out the rule-based classification stage that consumes it:

1. **Formal Rule-Based Classification**: Apply the literal, deterministic CAS decision rule (Elliptical / Spiral / Irregular, Section 3) to the verified feature table, rather than the qualitative narrative matching used for the Week 6 benchmarking discussion.
2. **Parameter Sensitivity / Robustness Sweep**: Systematically perturb the three decision thresholds (C_{thresh}, A_{thresh}, S_{thresh}) by $\pm 10\%$ in every combination and check whether the resulting class counts and accuracy are stable.
3. **Per-Galaxy Misclassification Analysis**: For every member where the rule and the NED/Galaxy Zoo ground truth disagree, identify which CAS parameter drove the disagreement and give a physical explanation.
4. **Consolidated Results Table**: Summarize detection counts, class counts, and accuracy for the JWST pipeline in one place, ready for the final project report.

All analysis in this report is deterministic and rule-based (fixed numeric threshold comparisons on the CAS table) – no machine learning or trained classifier is used anywhere in this step.

2 Parameter Sensitivity and Robustness Analysis

2.1 Methodology

The rule-based classifier assigns a class from the three CAS parameters as

$$\text{class}(C, A, S) = \begin{cases} \text{E} & C > C_{\text{thresh}} \text{ and } A < A_{\text{thresh}} \text{ and } S < S_{\text{thresh}} \\ \text{S} & C \leq C_{\text{thresh}} \text{ and } A \geq A_{\text{thresh}} \\ \text{Irr} & \text{otherwise} \end{cases} \tag{1}$$

with baseline thresholds $C_{\text{thresh}} = 2.5$, $A_{\text{thresh}} = 0.20$, $S_{\text{thresh}} = 0.10$ (Conselice 2003-consistent, carried over unchanged from Week 6). To test robustness, each threshold is independently perturbed by $\pm 10\%$ ($3 \times 3 \times 3 = 27$ combinations) and the rule is re-applied to the same five verified CAS rows from Table 2. This isolates sensitivity to *decision-boundary placement* from sensitivity to the underlying feature extraction (already validated in Week 6).

2.2 Results and Interpretation

Table 1: Robustness sweep: representative threshold combinations and the resulting class counts / accuracy over the 5 Stephan's Quintet members. The full 27-combination grid is included; only the extremes of the $\pm 10\%$ range are shown here for space.

C_{thresh}	A_{thresh}	S_{thresh}	N_E	N_S	N_{Irr}
2.25 (–10%)	0.18 (–10%)	0.09 (–10%)	0	5	0
2.50 (baseline)	0.20 (baseline)	0.10 (baseline)	0	5	0
2.75 (+10%)	0.22 (+10%)	0.11 (+10%)	0	5	0

Finding: all 27 threshold combinations in the $\pm 10\%$ grid produce the identical classification (N_E=0, N_S=5, N_{Irr}=0; accuracy fixed at $2/5 = 40.0\%$). At first glance this looks like an ideal, perfectly robust classifier – but

it is actually the opposite signal. None of the three thresholds sit anywhere near the measured C, A, S values for these five galaxies (Table 2: $C \in [1.67, 2.24]$, well short of the elliptical cutoff even before perturbation), so every member falls into the same branch of the decision rule regardless of a $\pm 10\%$ nudge. The sweep is therefore reporting *threshold insensitivity*, not *classification correctness* – the rule is saturated, not well-calibrated. This is the key actionable finding of this week's analysis and is discussed further in Section 4.

3 Finalized Rule-Based Classification (JWST Pipeline)

3.1 Verified Input Features

Table 2 reproduces the Week 6-verified CAS feature table (unchanged this week – feature extraction itself is not re-run) alongside the NED/Galaxy Zoo ground-truth class used for scoring.

Table 2: Week 6-verified CAS features (JWST) and NED/Galaxy Zoo ground truth.

Galaxy ID	C	A	S	NED Type	Ground Truth
NGC 7317	2.2404	0.6769	0.0000	E4	E
NGC 7318A	1.8756	0.2510	0.0089	E2 pec	Irr
NGC 7318B	1.7012	0.3066	0.0000	SB(s)bc pec	Irr
NGC 7319	1.6731	0.2329	0.0003	SB(s)cd pec	S
NGC 7320	1.7004	0.3897	0.0000	SA(s)d	S

3.2 Summary Statistics

Table 3: Final classification summary at baseline thresholds (C_{thresh}=2.5, A_{thresh}=0.20, S_{thresh}=0.10).

Metric	JWST Pipeline
Total galaxies scored (verified members)	5
Elliptical (E) count	0
Spiral (S) count	5
Irregular (Irr) count	0
Correct (matching ground truth)	2 / 5
Accuracy	**40.0%**

3.3 Per-Galaxy Misclassification Analysis

Table 4: Per-galaxy agreement between the rule-based prediction and ground truth.

Galaxy ID	GT	Pred.	Result	Likely Cause
NGC 7317	E	S	**Wrong**	A=0.6769 driven high by a superimposed foreground star; well above A_{thresh} despite a genuinely elliptical light profile.
NGC 7318A	Irr	S	**Wrong**	A=0.2510 clears A_{thresh} but is too low to separately flag as merger/irregular under this 3-way rule; tidal distortion registers as ordinary asymmetry.
NGC 7318B	Irr	S	**Wrong**	Same mechanism as NGC 7318A – the interacting pair's disturbed isophotes are not distinguished from a normal spiral by C, A, S alone at these thresholds.
NGC 7319	S	S	**Correct**	$C \leq C_{\text{thresh}}$ and $A \geq A_{\text{thresh}}$ – rule behaves as intended.
NGC 7320	S	S	**Correct**	Same – foreground spiral's moderate asymmetry correctly clears A_{thresh}.

Both misclassifications of a non-spiral member (NGC 7317 as E→S, and the two interacting members as Irr→S) share a common root cause: **the current 3-branch rule has no asymmetry *upper* bound**, so any object with

$A \geq A_{\text{thresh}}$ is labeled Spiral regardless of how extreme A actually is. NGC 7317's $A = 0.6769$ is $3.4\times$ higher than A_{thresh} – a value well into Conselice's merger/irregular range ($A > 0.35$) – yet the rule has no branch that would route it to Irr instead of S. This is a classification-rule gap, not a measurement error (Week 6 already validated the C/A/S extraction itself against independent sanity checks).

4 Conclusion and Next Steps

Week 7 converted Week 6's verified CAS features into a fully deterministic, rule-based classification and stress-tested it for robustness. Two concrete findings result:

1. The classifier is perfectly stable under $\pm 10\%$ threshold perturbation, but only because every member currently collapses into the same (Spiral) branch – a saturation artifact, not evidence that the decision boundaries are well placed.
2. 3 of 5 members are misclassified, and all three share the same failure mode: the rule has no upper asymmetry bound to separate ordinary Spirals from highly asymmetric Ellipticals/Irregulars.

4.1 Work Planned for Week 8

- **Decision-rule revision**: add an explicit upper-A branch (or a C–A–S decision surface informed by the actual measured value ranges in Table 2, rather than thresholds carried over unchanged since Week 3) so that high-A objects route to Irr/E as appropriate instead of defaulting to S.
- **Scale-up**: extend the validated pipeline to additional Hickson Compact Group targets beyond Stephan's Quintet, using the same reusable segmentation/CAS machinery.
- **Broader benchmarking**: re-run the accuracy and robustness checks of this report once the decision rule is revised, to confirm the fix actually improves accuracy rather than just shifting which branch the saturation falls into.

End Of Weekly Report